

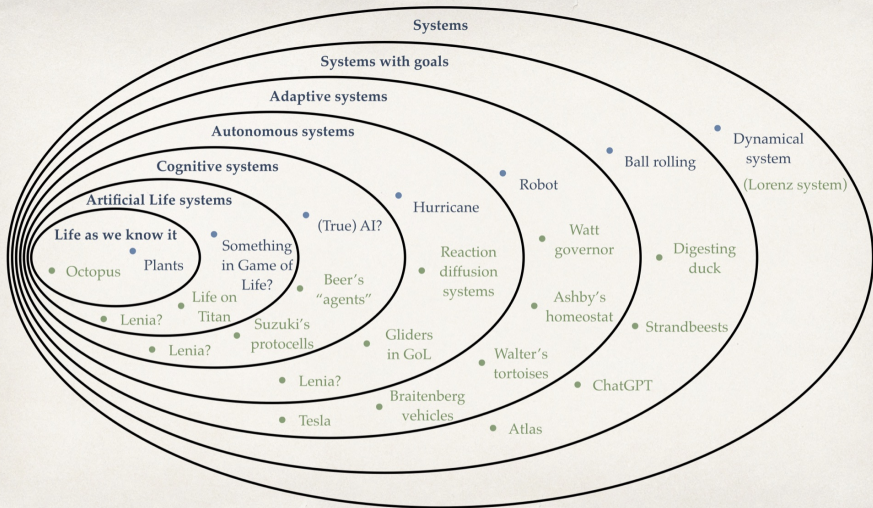
MINIMAL AGENCY

"Intelligent agency"

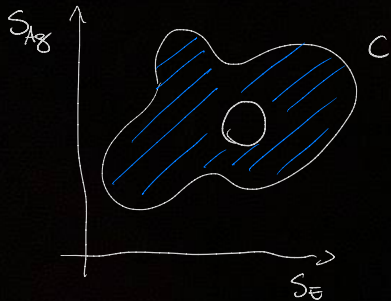
- human reasoning
- causal reasoning
- learning

Minimal agency
(artificial life?)
e.g. gliders, bacteria?

- no probabilities (for now)
 - no info theory
 - no (Pearl) causality
 - ...
- boundaries
- dynamical systems
- agent-env relations



An agent's constraints

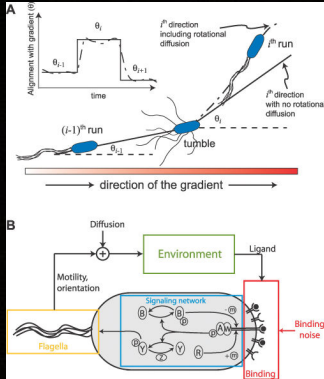


$$C \subset s_A \times s_E$$

... over time ($\sim$ viability theory)



Minimal notions of "models"



"perfect adaptation"

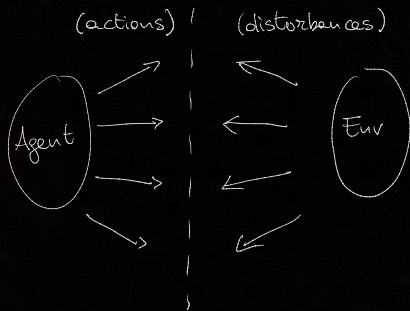


"integral control"

Good regulator "theorem"

"Every good regulator of a system must be a model of that system"

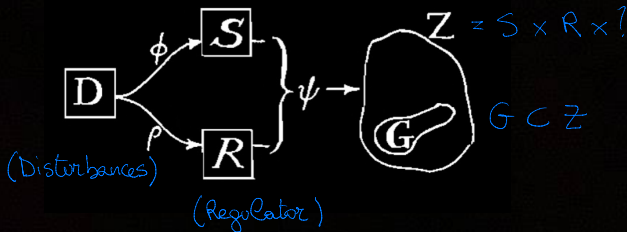
(Conant & Ashby, 1970)



(law of requisite variety)

The setup of the GR "T"

(System) Fig. 1



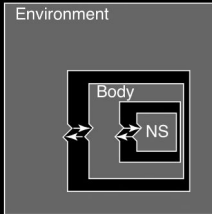
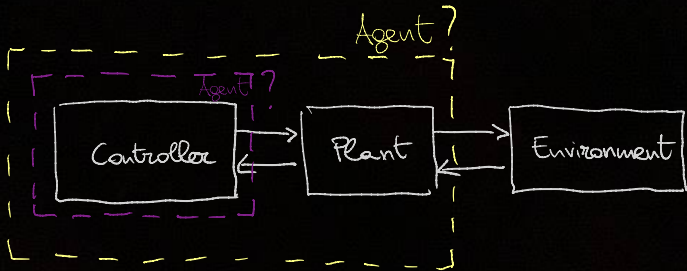
Issues with the GR "T"

- Confusing math. statement ("theorem")
 - set-theoretic setup
- No proof
 - probabilities for a "proof"
- No explicit use of dynamical systems
 - my problem with it

Solutions

- set-theory (internal model principle)
- probabilities (John?)

Internal model principle (setup)

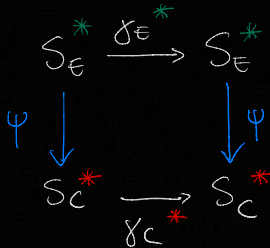


Internal model principle

IMP (sets, functions, relations)

$$\gamma_E : S_E \times S_P \rightarrow S_E$$

$$\gamma_C : S_C \times S_P \rightarrow S_C$$



Bayesian MP (?)

Bayesian inference

Ass. ψ is surjective

(powerset $\setminus \{\}$)

If $\psi: S_E^* \rightarrow S_C^*$ is surjective, $\exists g: S_C^* \rightarrow \mathcal{P}S_E^*$ s.t.

$$\begin{array}{ccc} S_C^* & \xrightarrow{\alpha_C^*} & \mathcal{P}S_C^* \\ g \downarrow & & \downarrow P_g \\ \mathcal{P}S_E^* & \xrightarrow{P_{\alpha_E^*}} & \mathcal{P}S_E^* \end{array}$$